SOURCE: MSC.141(76).pdf
should form part of the tests records.

The height of transverse or longitudinal bulkheads which are taken into account as effective to confine the assumed accumu lated sea water in the compartment concerned in the damaged ro-ro deck should be at least 4 m in height unless the height of water is less than 0.5 m. In such cases the height of the bulkhead may be calculated in accordance with the following:

Bh = 8hw

where Bh is the bulkhead height; and
hw is the height of water.

In any event, the minimum height of th e bulkhead should be not less than 2.2 m. However, in the case of a ship with hangi ng car decks, the minimum height of the bulkhead should be not less than the height to the underside of the hanging car deck when in its lowered position.

.5 In order to ensure that the model mo tion characteristics represent those of the actual ship it is important that the model is inclined in the intact condition so that the intact GM is verified. The mass dist ribution should be measured in air. The transverse radius of gyration of the actua l ship should be in the range 0.35B to

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SOURCE: MSC.141(76).pdf
the intact GM is verified. The mass dist ribution should be measured in air. The transverse radius of gyration of the actua l ship should be in the range 0.35B to 0.4B and the longitudinal radius of gyration should be in the range 0.2L to 0.25L.

Note: While inclining and rolling the model in the damage condition may be accepted as a check for the purpose of verifying the residual stability curve, such tests should not be accepted in lieu of the intact tests.

.6 It is assumed that the ventila tors of the damaged compartment of the actual ship are adequate for unhindered flooding and m ovement of the floodwater. However in trying to scale down the ventilating a rrangements of the actual ship undesirable scale effects may be introduced. In order to ensure that these do not occur it is

recommended to construct the ventilating arrangements to a larger scale than that of the model, ensuring that this does not affect the flow of water on the car deck.

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SOURCE: MSC.141(76).pdf

recommended to construct the ventilating arrangements to a larger scale than that of the model, ensuring that this does not affect the flow of water on the car deck.

.7 It is deemed appropriate to consider a damage shape representative of a cross section of the striking ship in the bow region. The 15° angle is based on a study of the cross section at a distance of B/5 from the bow for a representative selection of vessels of different types and sizes.

The isosceles triangular profile of the prismatic damage shape is that corresponding to the load waterline.

Additionally, in cases where side casings of width less than B/5 are fitted and in order to avoid any possible scale effects, the damage length in way of the side casings should not be less than 25 mm.

3.3 In the original model test method of resolution 14 of the 1995 SOLAS Conference the effect of heeling induced by the maximu m moment deriving from any of passenger crowding, launching of survival craft, wind and turning was not considered even though this effect was part of SOLAS. Results from an investigation have shown however that it

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SOURCE: MSC.141(76).pdf
crowding, launching of survival craft, wind and turning was not considered even though this effect was part of SOLAS. Results from an investigation have shown however that it would be prudent to take these effects into account and to retain the minimum of 1° heel towards the damage for practical purposes. It is to be noted that heeling due to turning was considered not to be relevant.

3.4 In cases where there is a margin in GM in the actual loading conditions compared to the GM limiting curve (derived from SOLAS 90) , the Administration may accept that this margin is taken advantage of in the model test. In such cases the GM limiting curve should be adjusted. This adjustment can be done as follows:

$$d = dS – 0.6 (dS - dLS)$$

where: d_S is the subdivision draught; and
d_{LS} is the lightship draught.

The adjusted curve is a straight line between the GM used in the model test at the subdivision draught and the intersection of the original SOLAS 90 curve and draught d.

RESOLUTION MSC.141(76)
(adopted on 5 December 2002)
REVISED MODEL TEST METHOD UNDER RESOLUTION 14
OF THE 1995 SOLAS CONFERENCEMSC 76/23/Add.1

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SOURCE: MSC.141(76).pdf
RESOLUTION MSC.141(76)
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ANNEX 17
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Paragraph 4 - Procedure for experiments

4.1 Wave spectra

The JONSWAP spectrum should be used as th is describes fetch and duration in limited seas, which correspond to the majority of the c onditions worldwide. In this respect it is important that not only the peak period of the wa ve train is verified but also that the zero crossing period is correct.

It is required that for every test run th e wave spectrum be recorded and documented. Measurements for this recording should be ta ken at the probe closest to the wave making machine.

It is also required that the model be instrumented so that its motions (roll, heave and pitch) as well as its attitude (heel, sinkage and trim) are monitored and recorded.

It has been found that it is not practical to set absolute limits for significant wave heights, peak periods and zero crossing periods of the model wave spectra, therefore an acceptable margin has been introduced.

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SOURCE: MSC.141(76).pdf
It has been found that it is not practical to set absolute limits for significant wave heights, peak periods and zero crossing periods of the model wave spectra, therefore an acceptable margin has been introduced.

4.2 To avoid interference of the mooring syst em with the ship dynamics, the towing carriage (to which the mooring system is attached) should follow the model at its actual drifting speed. In a sea state with irregular waves the drift speed will not be constant; a constant carriage speed would result in low frequency, large amplitude drift oscillations, which may affect the model behaviour.

4.3 A sufficient number of tests in different wave trains are necessary to ensure statistical reliability, i.e. the objective is to determin e with a high degree of confidence that an unsafe ship will capsize in the selected c onditions. A minimum number of 10 runs are considered to provide a reasonable level of reliability.

Paragraph 5 - Survival criteria

The contents of this paragraph are considered self-explanatory.

Paragraph 6 - Test approval

The following documents should be part of the report to the Administration:

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SOURCE: MSC.141(76).pdf
Paragraph 5 - Survival criteria

The contents of this paragraph are considered self-explanatory.

Paragraph 6 - Test approval

The following documents should be part of the report to the Administration:

.1 damage stability calculations for worst SOLAS and midship damage (if different);

.2 general arrangement drawing of the mode l together with details of construction and instrumentation;

.3 inclining experiment and measurements of radii of gyration;

.4 nominal and measured wave spectra (at the 3 different locations for a